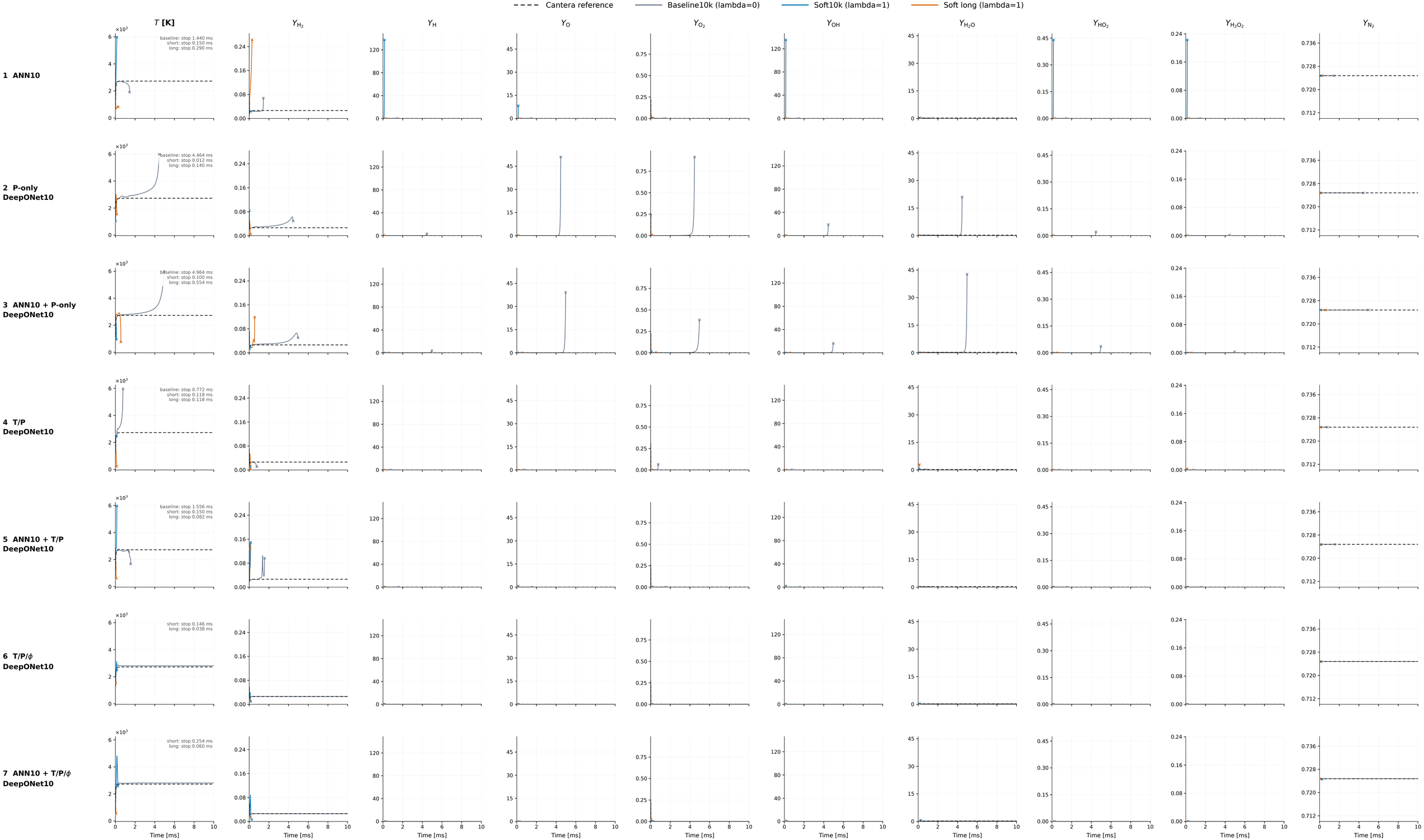
Lambda 1 continuation: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=1) · Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 04 | Cantera vs NN NRMSE

T0=1317.5 K P0=1.244 atm phi0=1.984 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

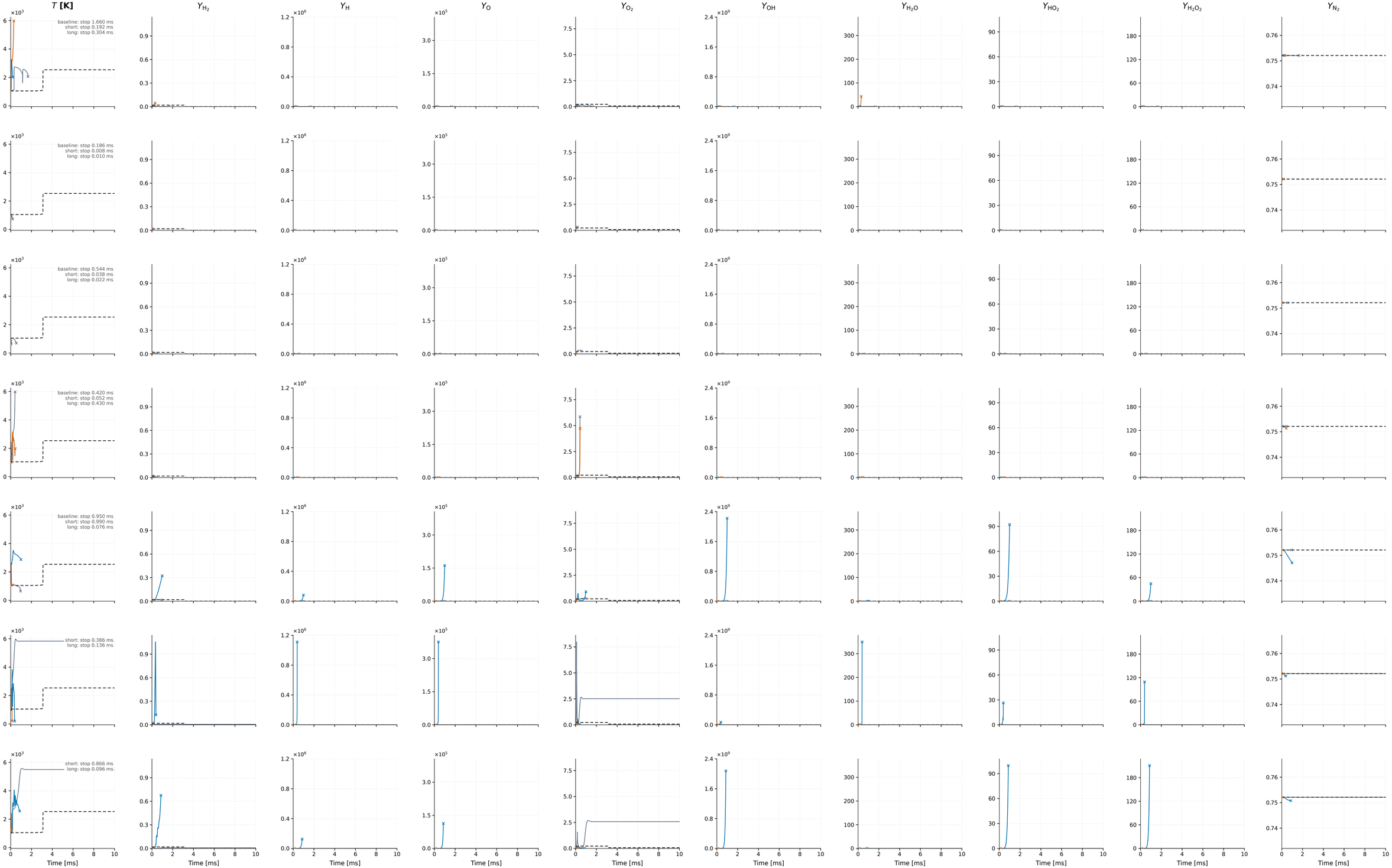
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.440 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.290 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 4.464 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.140 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 4.964 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.100 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.554 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.772 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.118 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.118 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.556 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.082 ms
6 T/P/ϕ DeepONet10	base10k	0.0309	0.0188	0.215	0.559	0.164	0.406	0.0425	0.406	0.363	6.13e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.146 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0296	0.0176	0.223	0.634	0.106	0.427	0.047	0.294	0.389	1.53e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.254 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 05 | Cantera vs NN NRMSE

T0=1049.0 K P0=4.666 atm phi0=0.674 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.660 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.192 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.304 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.010 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.544 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.022 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.420 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.052 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.430 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.950 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.990 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
6 T/P/ϕ DeepONet10	base10k	1.71	0.835	29.8	1.22e+03	16.9	336	0.817	2.37	0.961	1.48e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.386 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	1.5	0.903	68.8	2.19e+03	16.8	203	0.602	2.85	0.951	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.866 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.096 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K]

baseline: stop 2.406 ms
short: stop 0.270 ms
long: stop 0.248 ms

baseline: stop 0.076 ms
short: stop 0.004 ms
long: stop 0.002 ms

baseline: stop 0.164 ms
short: stop 0.004 ms
long: stop 0.026 ms

baseline: stop 0.028 ms
short: stop 0.058 ms
long: stop 0.214 ms

baseline: stop 0.078 ms
short: stop 1.322 ms
long: stop 0.122 ms

baseline: stop 0.018 ms
short: stop 0.004 ms
long: stop 0.152 ms

baseline: stop 0.078 ms
short: stop 1.044 ms
long: stop 0.104 ms

Y_{H_2} Y_H Y_O Y_{O_2} Y_{OH} Y_{H_2O} Y_{HO_2} $Y_{H_2O_2}$ Y_{N_2}

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 07 | Cantera vs NN NRMSE

T0=930.6 K P0=3.722 atm phi0=0.754 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

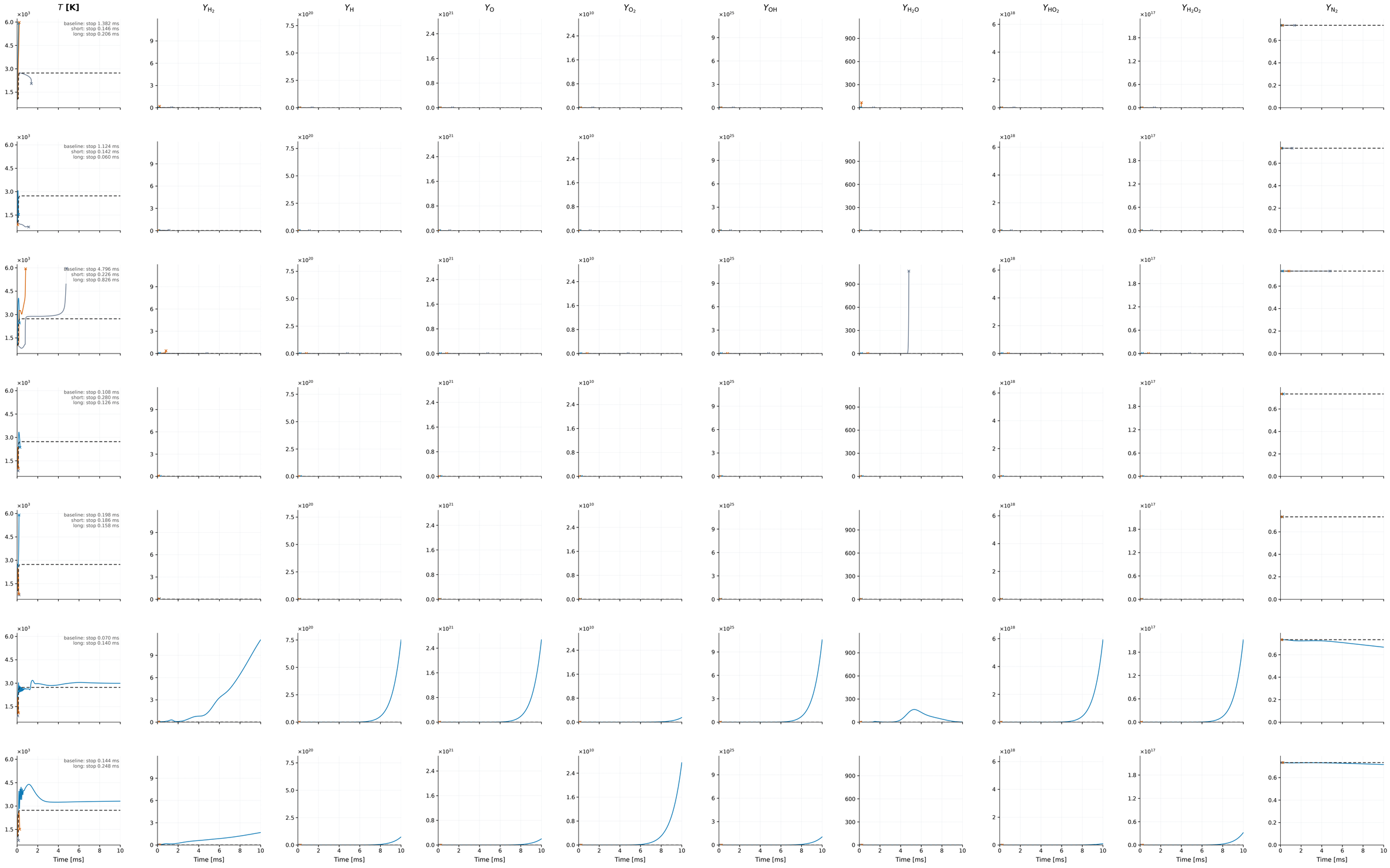
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.406 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.270 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.248 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.002 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.164 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.214 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.322 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.004 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.152 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.044 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.104 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 18 | Cantera vs NN NRMSE

T0=1069.1 K P0=2.170 atm phi0=1.544 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.382 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.146 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.206 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.124 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.142 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 4.796 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.226 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.826 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.126 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.198 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.158 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
6 T/P/ϕ DeepONet10	soft10k	0.101	292	1.81e+23	1.09e+24	1.24e+10	3.21e+27	282	8.47e+22	3.71e+22	0.0429	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.140 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.144 ms
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.286	56.1	1.82e+22	8.33e+22	2.41e+11	3.26e+26	0.991	1.43e+21	5.75e+21	0.0105	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.248 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference | Baseline10k (lambda=0) | Soft10k (lambda=1) | Soft long (lambda=1)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	baseline: stop 0.838 ms short: stop 0.180 ms long: stop 0.186 ms									
2 P-only DeepONet10	baseline: stop 1.052 ms short: stop 0.012 ms long: stop 0.076 ms									
3 ANN10 + P-only DeepONet10	baseline: stop 2.922 ms short: stop 0.024 ms long: stop 0.152 ms									
4 T/P DeepONet10	short: stop 0.058 ms long: stop 0.158 ms									
5 ANN10 + T/P DeepONet10	short: stop 0.282 ms long: stop 0.136 ms									
6 T/P/ϕ DeepONet10	short: stop 0.066 ms long: stop 0.098 ms									
7 ANN10 + T/P/ϕ DeepONet10	short: stop 0.080 ms long: stop 0.166 ms									

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 32 | Cantera vs NN NRMSE

T0=1140.7 K P0=3.354 atm phi0=1.463 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

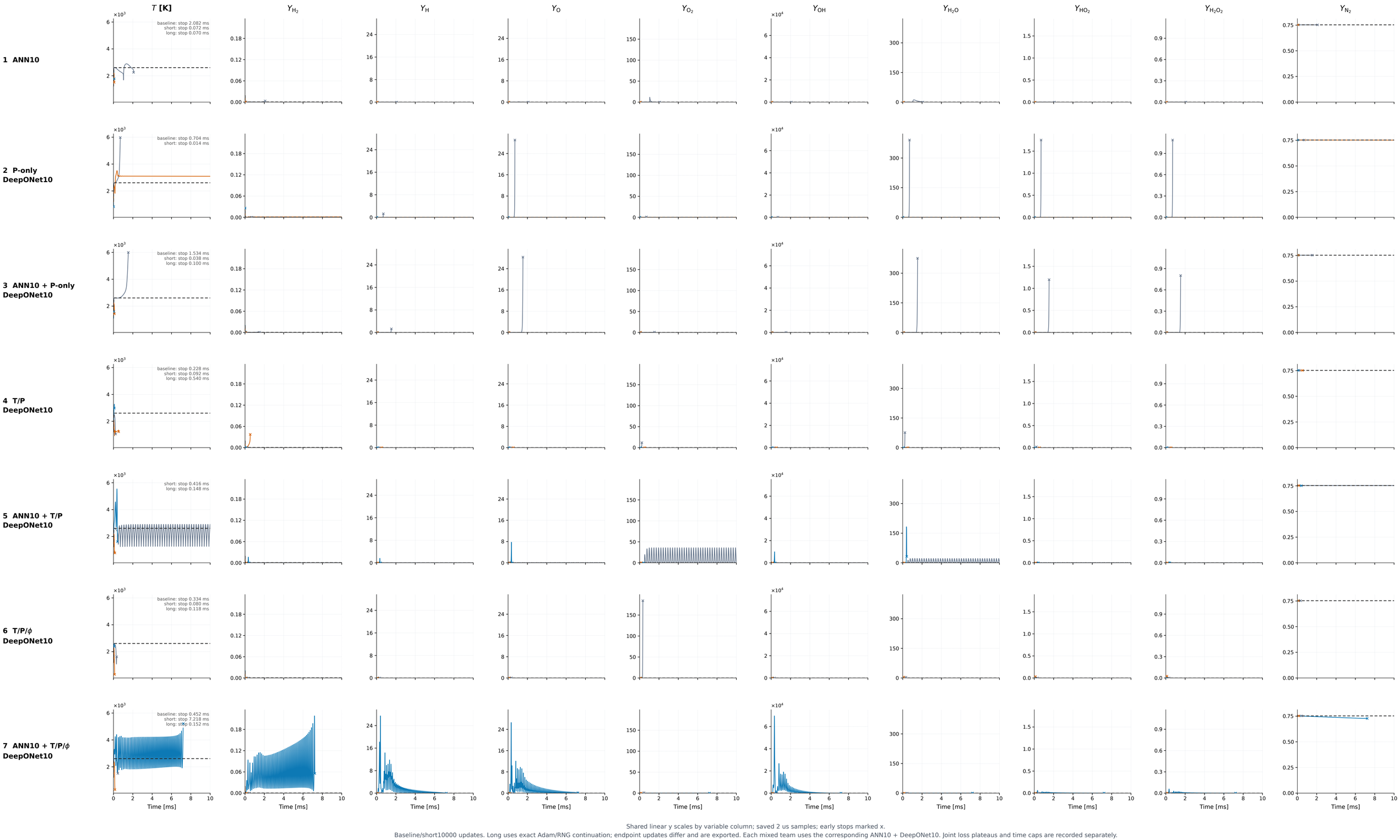
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.838 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.186 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.052 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.076 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.922 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.152 ms
4 T/P DeepONet10	base10k	0.0202	0.0859	0.158	0.382	0.588	0.0956	0.0328	1.22	0.997	1.06e-15	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.158 ms
5 ANN10 + T/P DeepONet10	base10k	0.0163	0.0579	0.204	0.378	0.395	0.172	0.0257	1.12	1.1	1.06e-15	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.282 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
6 T/P/ϕ DeepONet10	base10k	1.07	1.94	2.39	2.27e+03	58	3.86e+03	0.953	23.8	421	1.06e-15	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.066 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.098 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.224	2.05	102	0.727	0.326	0.829	75.3	1.85	30.6	1.06e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 39 | Cantera vs NN NRMSE

T0=1250.1 K P0=2.297 atm phi0=0.646 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

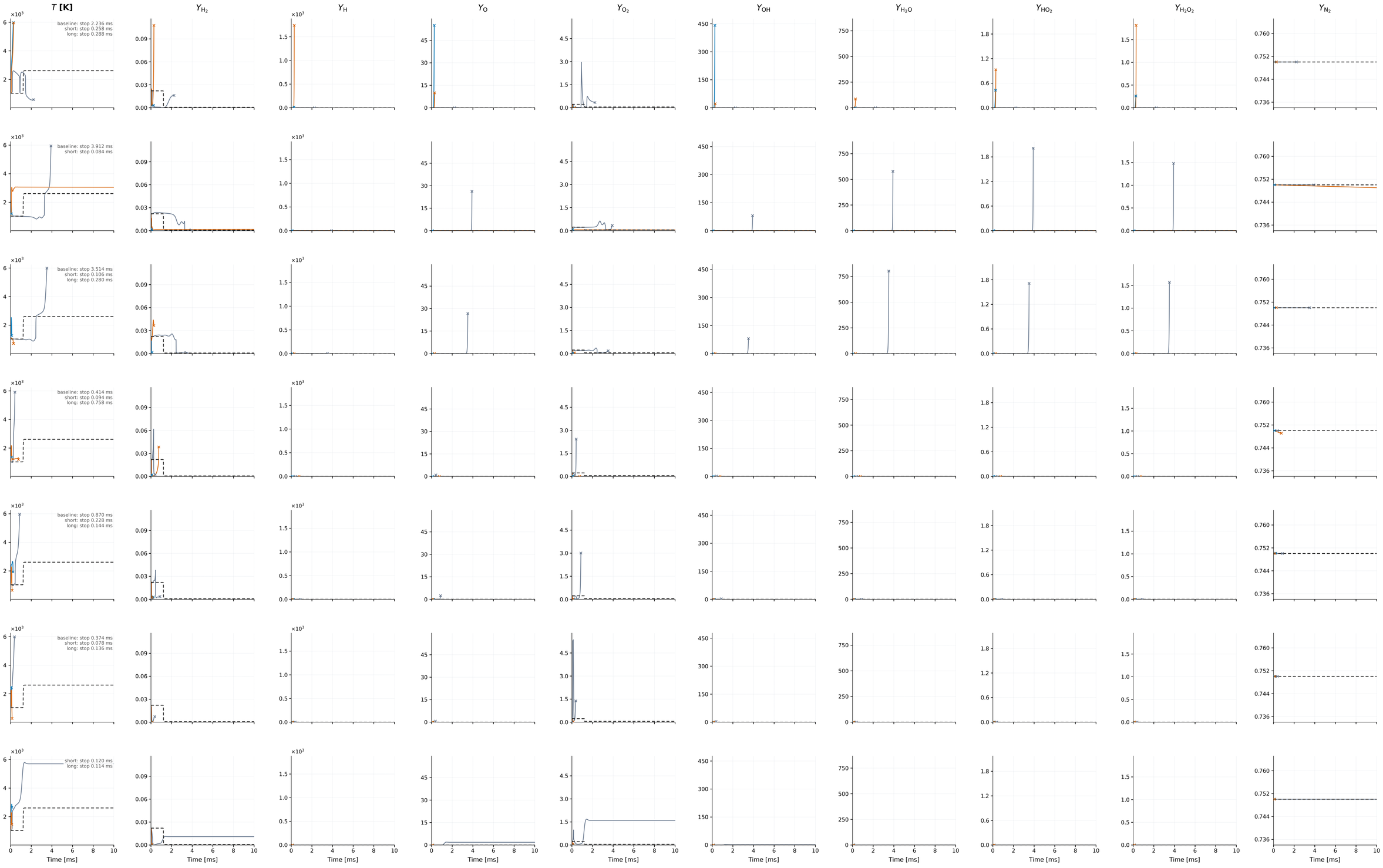
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.082 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.072 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.704 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.014 ms
2 P-only DeepONet10	long	0.19	0.557	3.69	2.06	0.315	1.38	0.0833	1.59	0.952	7.83e-04	Complete
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.534 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.100 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.228 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.092 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.540 ms
5 ANN10 + T/P DeepONet10	base10k	0.225	0.427	126	12	214	167	67.7	97.8	29.2	5.90e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.416 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.334 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.118 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.452 ms
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 7.218 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.152 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 44 | Cantera vs NN NRMSE

T0=1031.1 K P0=2.272 atm phi0=0.770 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

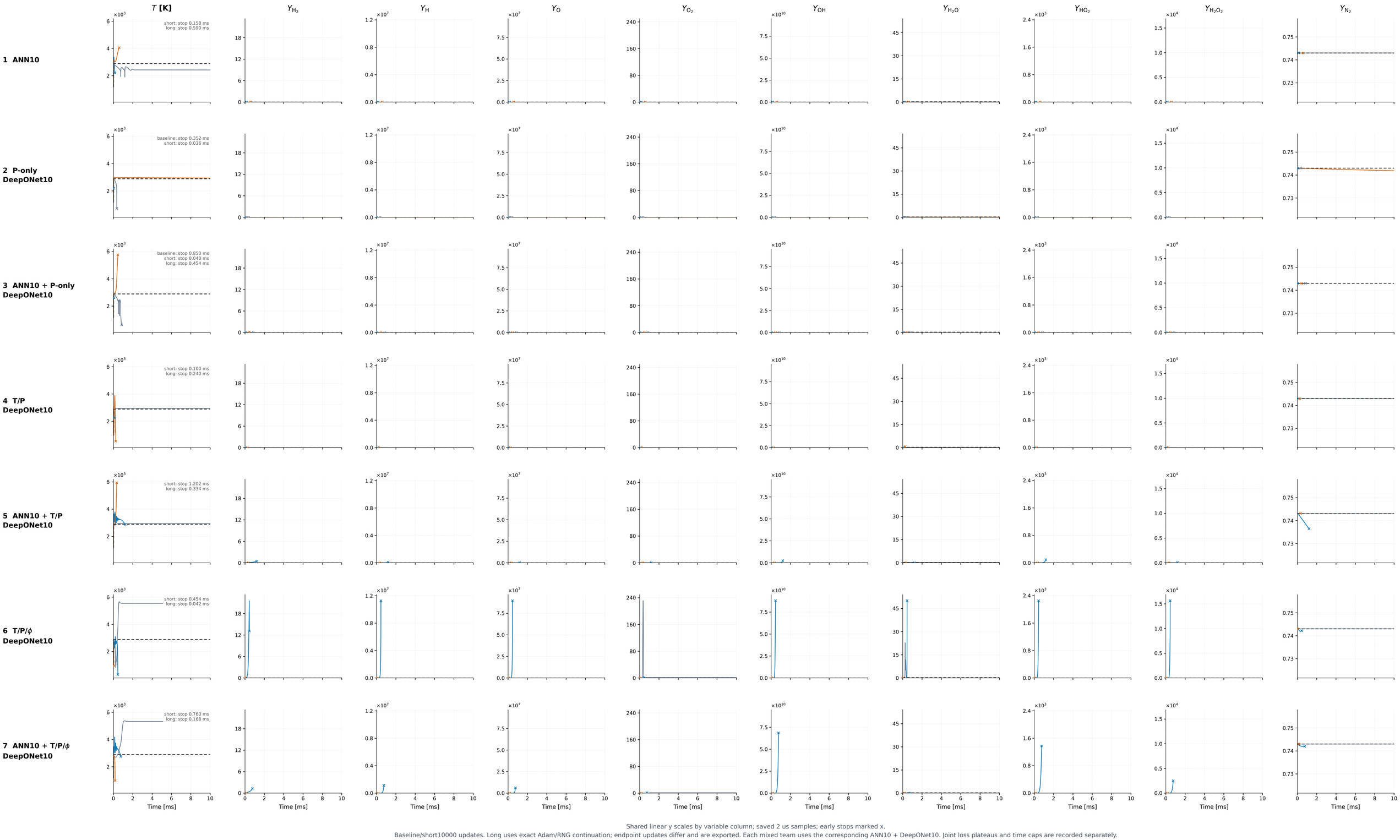
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.236 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.258 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.288 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 3.912 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.084 ms
2 P-only DeepONet10	long	0.321	0.908	4.51	2.67	0.628	1.55	0.336	1.04	0.978	7.60e-04	Complete
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 3.514 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.106 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.414 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.094 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.758 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.870 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.228 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.144 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.374 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	1.21	1.51	31.7	859	15.7	160	0.333	6.13	5.11	2.22e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.120 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.114 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 45 | Cantera vs NN NRMSE

T0=1181.9 K P0=4.842 atm phi0=1.099 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

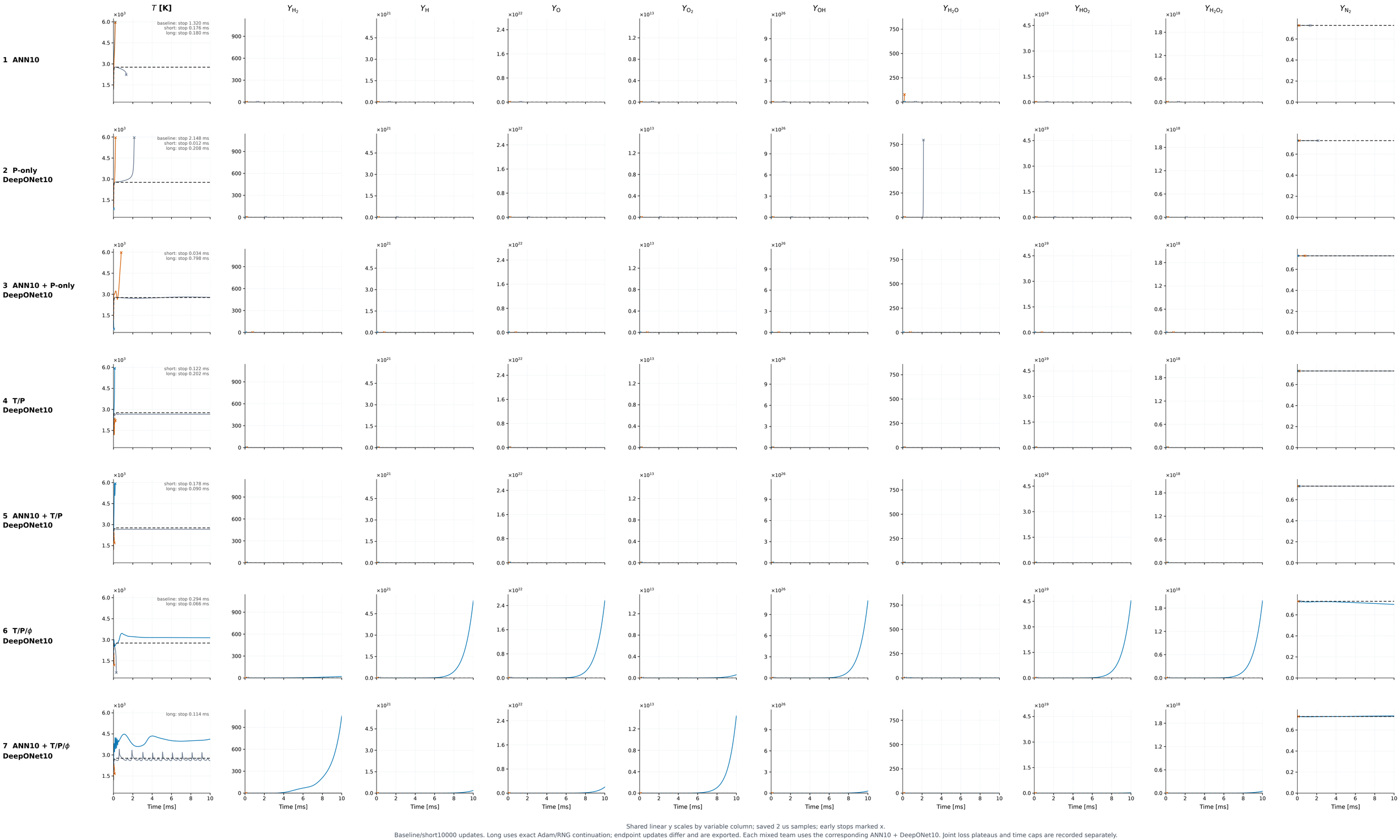
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.154	1.63	0.954	0.964	0.621	0.93	0.638	0.705	0.788	1.49e-16	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.158 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.590 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
2 P-only DeepONet10	long	0.0282	0.322	0.281	0.208	0.445	0.082	0.165	0.891	0.791	9.15e-04	Complete
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.850 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.040 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.454 ms
4 T/P DeepONet10	base10k	0.0217	0.262	0.261	0.199	0.511	0.0576	0.0293	0.84	0.605	1.49e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.100 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.240 ms
5 ANN10 + T/P DeepONet10	base10k	0.0214	0.315	0.312	0.16	0.505	0.116	0.0315	0.885	0.653	1.49e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.202 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.334 ms
6 T/P/ϕ DeepONet10	base10k	0.897	2.9	2.57	818	1.26e+03	788	5.08	31.5	155	1.49e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.454 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.042 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.805	3.6	3.74	1.03e+03	80.1	212	0.391	12.5	67.8	1.49e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.760 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.168 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 53 | Cantera vs NN NRMSE

T0=1257.8 K P0=2.103 atm phi0=1.799 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.320 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.176 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.148 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
3 ANN10 + P-only DeepONet10	base10k	0.0124	0.292	0.396	0.464	0.0498	0.204	0.136	0.121	0.506	7.62e-16	Complete
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.034 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.798 ms
4 T/P DeepONet10	base10k	0.0377	0.0241	0.286	0.477	0.312	0.332	0.0163	0.592	0.456	7.62e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.202 ms
5 ANN10 + T/P DeepONet10	base10k	0.0356	0.0124	0.425	0.598	0.201	0.426	0.106	0.415	0.373	7.62e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.178 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.090 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.294 ms
6 T/P/ϕ DeepONet10	soft10k	0.147	360	9.33e+23	1.08e+25	1.37e+13	3.34e+28	1.32	1.87e+24	2.05e+24	0.0195	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.066 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.05	0.249	0.583	0.893	0.258	0.86	0.241	0.547	1.1	7.62e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.465	1.17e+04	3.18e+22	8.21e+23	3.06e+14	8.90e+26	0.925	8.46e+21	4.97e+22	4.06e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.114 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference | Baseline10k (lambda=0) | Soft10k (lambda=1) | Soft long (lambda=1)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	baseline: stop 2.010 ms short: stop 0.092 ms long: stop 1.170 ms									
2 P-only DeepONet10	baseline: stop 0.524 ms short: stop 0.016 ms									
3 ANN10 + P-only DeepONet10	baseline: stop 0.954 ms short: stop 0.024 ms									
4 T/P DeepONet10	baseline: stop 0.240 ms short: stop 1.832 ms long: stop 0.180 ms									
5 ANN10 + T/P DeepONet10	short: stop 1.100 ms									
6 T/P/ϕ DeepONet10	baseline: stop 0.452 ms short: stop 0.080 ms long: stop 0.218 ms									
7 ANN10 + T/P/ϕ DeepONet10	short: stop 0.774 ms									

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 57 | Cantera vs NN NRMSE

T0=1412.0 K P0=4.083 atm phi0=0.677 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

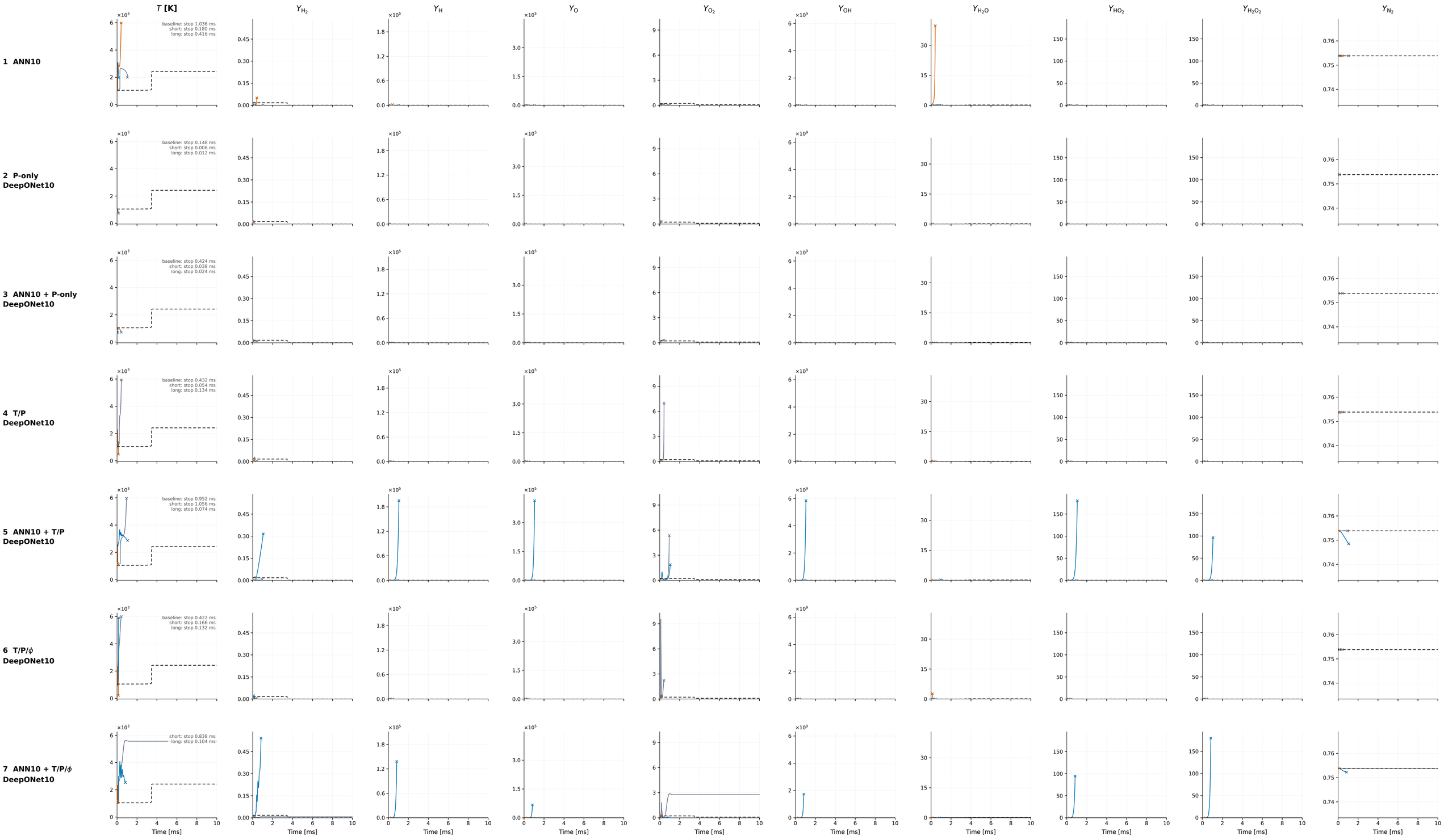
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.010 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.092 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 1.170 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.524 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
2 P-only DeepONet10	long	0.0287	0.458	1.09	0.477	0.24	0.175	0.119	0.41	0.375	1.02e-03	Complete
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.954 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
3 ANN10 + P-only DeepONet10	long	0.0448	0.344	0.549	0.579	0.25	0.303	0.229	0.145	0.142	6.87e-04	Complete
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.240 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.832 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
5 ANN10 + T/P DeepONet10	base10k	0.332	0.813	56.4	0.932	63.3	136	59.5	57.4	32.6	1.48e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.100 ms
5 ANN10 + T/P DeepONet10	long	0.202	4.93	32.3	35.1	0.673	788	0.953	34.8	18.1	8.40e-03	Complete
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.452 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.218 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.977	9.31	18.2	665	32.5	95.2	0.146	6.37	26.8	1.48e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.774 ms
7 ANN10 + T/P/ϕ DeepONet10	long	0.217	0.559	7.37	10.6	1.89	79.5	0.819	18.7	5.67	7.83e-03	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 64 | Cantera vs NN NRMSE

T0=1045.3 K P0=4.301 atm phi0=0.595 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

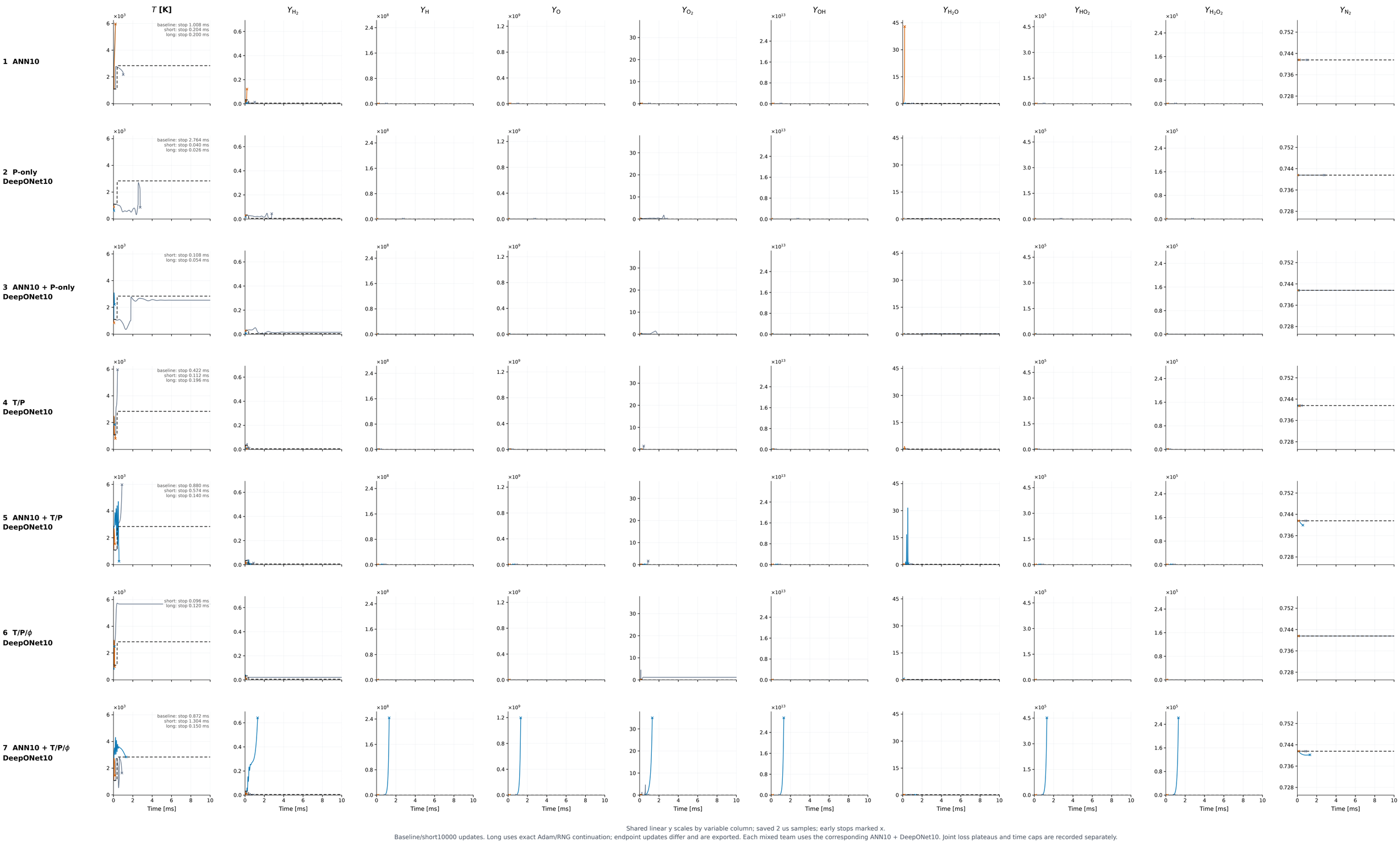
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.036 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.180 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.416 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.148 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.006 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.424 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.024 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.432 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.054 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.952 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.056 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.422 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.132 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	1.72	0.836	120	3.17e+03	16.6	271	0.621	3.01	0.949	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.838 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.104 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 68 | Cantera vs NN NRMSE

T0=1086.3 K P0=3.702 atm phi0=1.168 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.008 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.204 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.200 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.764 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.040 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
3 ANN10 + P-only DeepONet10	base10k	0.295	1.67	0.514	0.978	5.87	0.866	0.624	2.66	163	4.49e-16	Complete
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.054 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.422 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.112 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.196 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.880 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.574 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.140 ms
6 T/P/ϕ DeepONet10	base10k	1.01	1.81	2.29	1.08e+03	26.3	575	0.851	7.61	18.2	4.49e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.096 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.120 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.872 ms
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.304 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

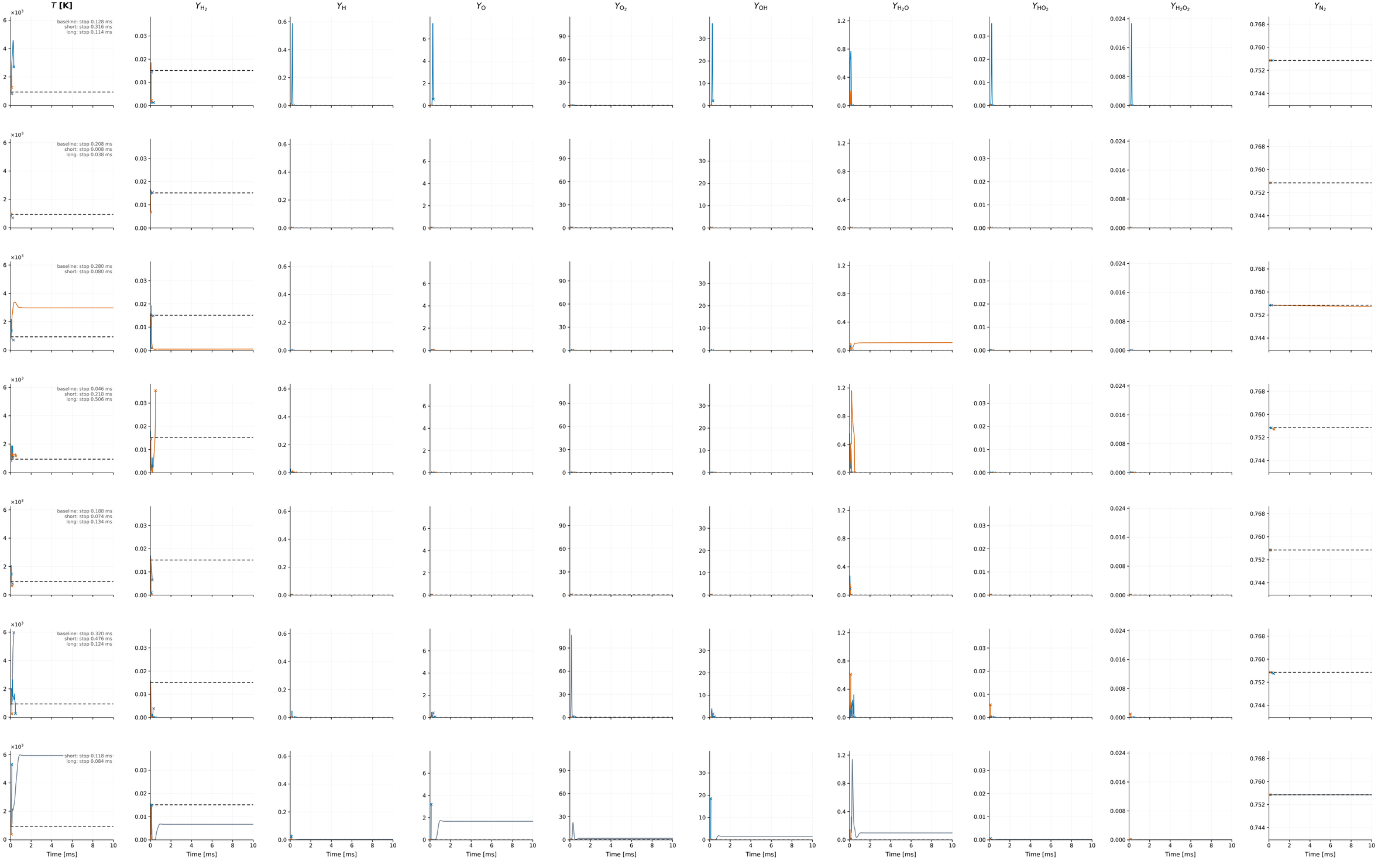
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 69 | Cantera vs NN NRMSE

T0=935.0 K P0=2.045 atm phi0=0.523 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.128 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.316 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.114 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.008 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.080 ms
3 ANN10 + P-only DeepONet10	long	2.18	0.959	1.30e+08	1.62e+09	0.459	7.81e+09	1.58e+06	218	17.3	3.49e-04	Complete
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.218 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.506 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.188 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.320 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.476 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	5.19	0.582	9.65e+08	1.69e+11	12.3	4.23e+11	2.17e+06	829	349	1.03e-15	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.118 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.084 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference | Baseline10k (lambda=0) | Soft10k (lambda=1) | Soft long (lambda=1)

	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}
1 ANN10	baseline: stop 0.918 ms short: stop 0.136 ms long: stop 0.208 ms									
2 P-only DeepONet10	baseline: stop 3.434 ms short: stop 0.098 ms long: stop 0.224 ms									
3 ANN10 + P-only DeepONet10	short: stop 0.210 ms long: stop 0.280 ms									
4 T/P DeepONet10	short: stop 0.340 ms long: stop 0.170 ms									
5 ANN10 + T/P DeepONet10	short: stop 0.178 ms long: stop 0.150 ms									
6 T/P/ϕ DeepONet10	baseline: stop 0.312 ms short: stop 0.324 ms long: stop 6.886 ms									
7 ANN10 + T/P/ϕ DeepONet10	short: stop 0.150 ms long: stop 0.490 ms									

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 73 | Cantera vs NN NRMSE

T0=1106.0 K P0=1.911 atm phi0=1.915 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.918 ms
1 ANN10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
1 ANN10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
2 P-only DeepONet10	**base10k**	—	—	—	—	—	—	—	—	—	—	Stop 3.434 ms
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.098 ms
2 P-only DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.224 ms
3 ANN10 + P-only DeepONet10	**base10k**	0.06	0.5	0.694	2.01	0.0571	0.946	0.199	0.207	0.307	7.64e-16	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
3 ANN10 + P-only DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.280 ms
4 T/P DeepONet10	**base10k**	0.0213	0.271	0.332	0.788	0.495	0.316	0.107	1.23	1.2	7.64e-16	Complete
4 T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.340 ms
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.170 ms
5 ANN10 + T/P DeepONet10	**base10k**	0.0125	0.0563	0.39	0.725	0.384	0.262	0.213	0.953	1.05	7.64e-16	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.178 ms
5 ANN10 + T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
6 T/P/ϕ DeepONet10	**base10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.312 ms
6 T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.324 ms
6 T/P/ϕ DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 6.886 ms
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0314	0.167	0.638	1.09	0.399	0.587	0.172	0.993	1.07	7.64e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
7 ANN10 + T/P/ϕ DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.490 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

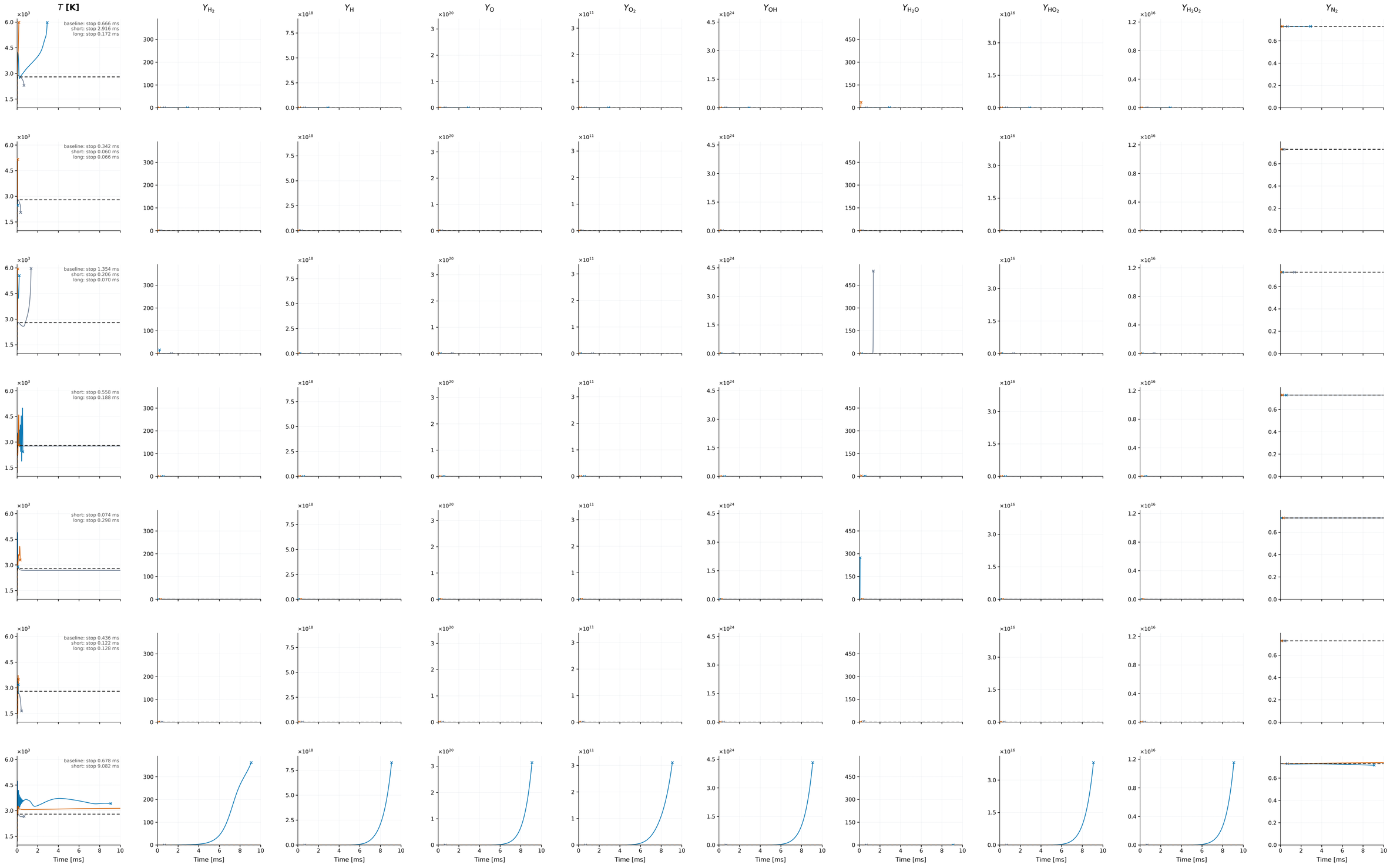
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 74 | Cantera vs NN NRMSE

T0=1222.9 K P0=4.971 atm phi0=1.766 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

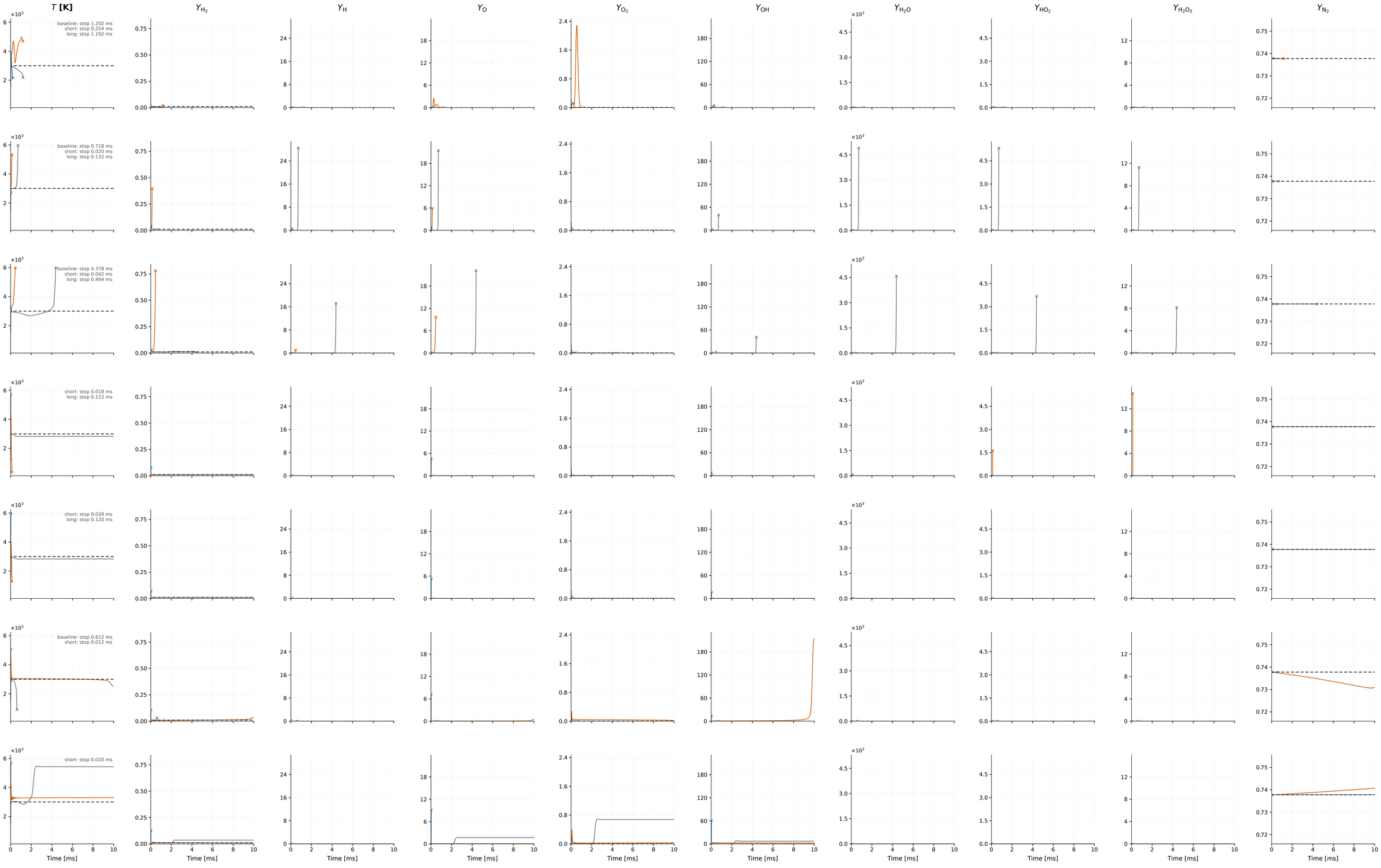
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.666 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 2.916 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.172 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.342 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.060 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.066 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.354 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.206 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
4 T/P DeepONet10	base10k	0.0146	0.0312	0.19	0.55	0.598	0.207	0.0463	1.03	0.965	1.83e-15	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.558 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.188 ms
5 ANN10 + T/P DeepONet10	base10k	0.0414	0.092	0.578	0.866	0.539	0.599	0.208	1.07	0.987	1.37e-15	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.298 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.436 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.128 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.678 ms
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 9.082 ms
7 ANN10 + T/P/ϕ DeepONet10	long	0.111	1.46	7.53	118	0.814	2.66e+03	0.951	45.9	99.8	8.35e-03	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Raw self-fed rollout, $\Delta t = 2\,\mu s$





Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 76 | Cantera vs NN NRMSE

T0=1523.6 K P0=3.626 atm phi0=1.349 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.202 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.204 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 1.192 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.718 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.132 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 4.378 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.042 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.464 ms
4 T/P DeepONet10	base10k	0.0553	0.0153	0.391	0.634	0.466	0.448	0.0778	0.653	0.534	6.02e-16	Complete
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
5 ANN10 + T/P DeepONet10	base10k	0.0546	8.51e-03	0.396	0.647	0.485	0.475	0.0889	0.671	0.58	6.02e-16	Complete
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.120 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.612 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.012 ms
6 T/P/ϕ DeepONet10	long	0.0392	0.473	1.64	31.1	5.38	1.58e+03	0.703	38.2	69.5	5.65e-03	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.714	2.11	1.21	590	89.7	351	0.664	27.1	163	6.02e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
7 ANN10 + T/P/ϕ DeepONet10	long	0.0991	0.144	1.11	16.2	5.03	143	0.803	30.2	19.3	2.21e-03	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Raw self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

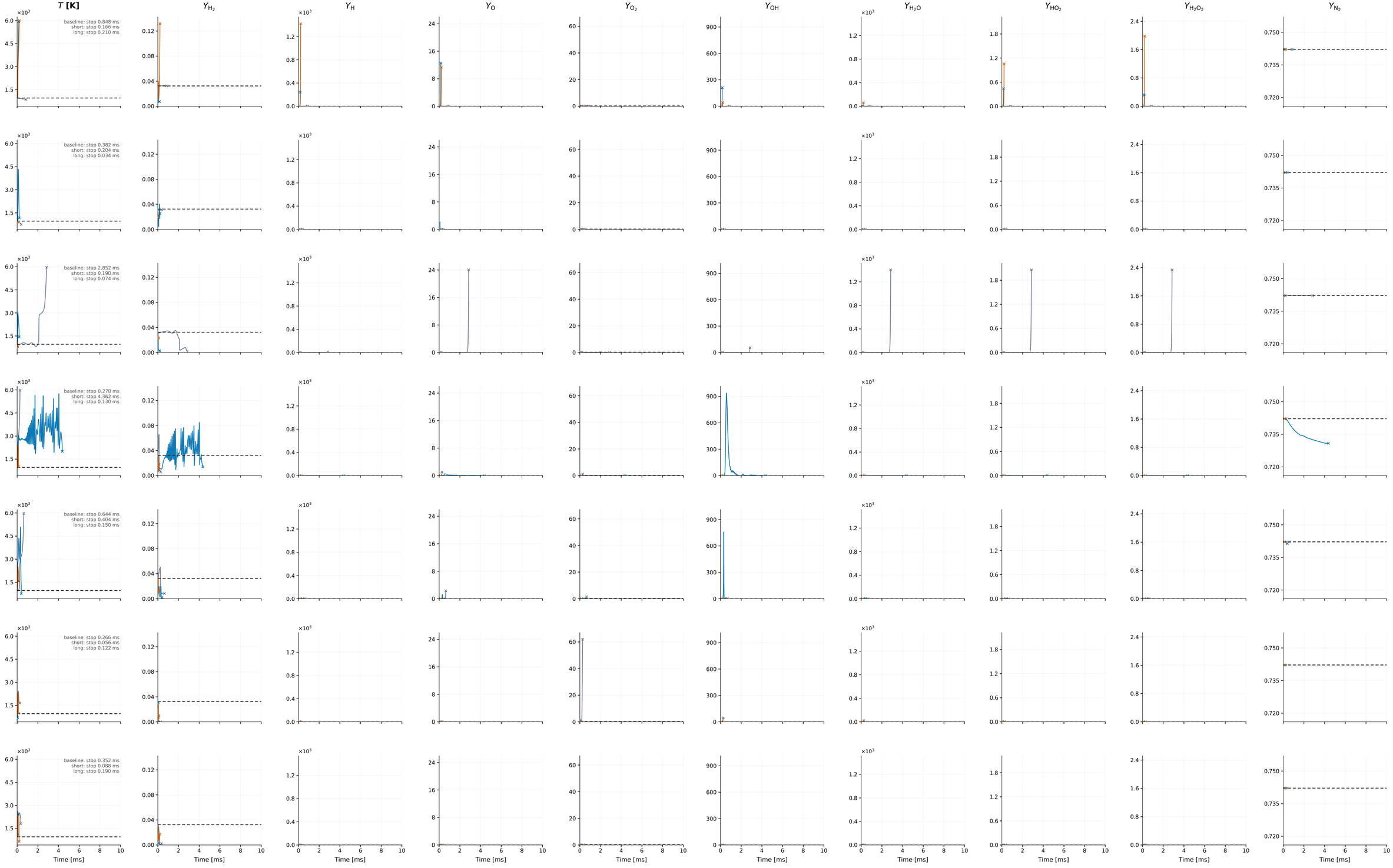
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 78 | Cantera vs NN NRMSE

T0=962.3 K P0=2.443 atm phi0=1.140 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.848 ms
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.166 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.210 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.382 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.204 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.034 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 2.852 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.190 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.278 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 4.362 ms
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.130 ms
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.644 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.404 ms
5 ANN10 + T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.150 ms
6 T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.266 ms
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.056 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.122 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.352 ms
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.088 ms
7 ANN10 + T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.190 ms

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 1 continuation: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Raw self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=1) Soft long (lambda=1)

1 ANN10

2 P-only DeepONet10

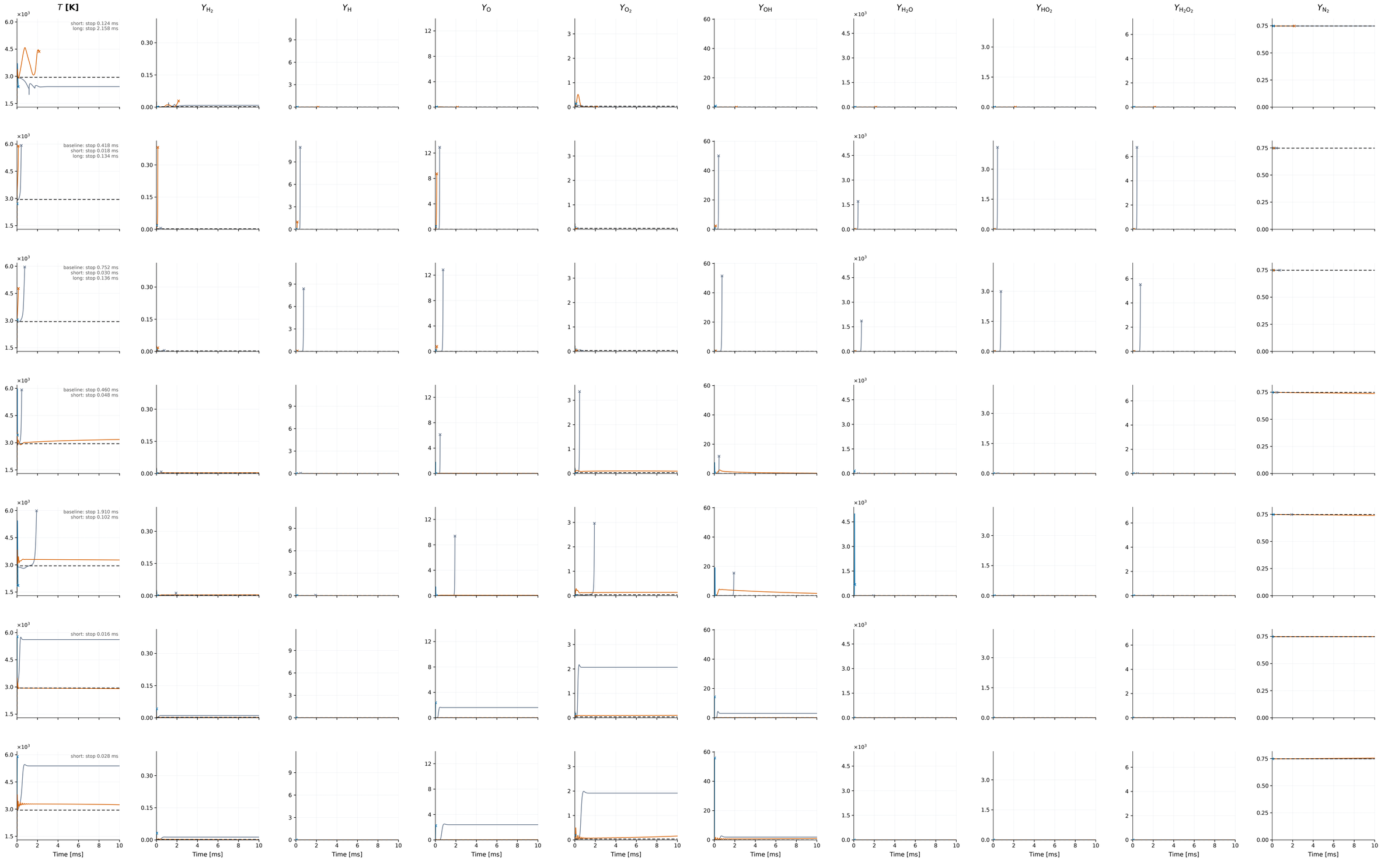
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 1 | Test IC 88 | Cantera vs NN NRMSE

T0=1502.1 K P0=4.980 atm phi0=0.859 | Raw self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.165	2.68	0.929	0.95	0.951	0.9	0.71	0.956	0.948	8.90e-16	Complete
1 ANN10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
1 ANN10	long	—	—	—	—	—	—	—	—	—	—	Stop 2.158 ms
2 P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.418 ms
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.018 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
3 ANN10 + P-only DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.752 ms
3 ANN10 + P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.030 ms
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.136 ms
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 0.460 ms
4 T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.048 ms
4 T/P DeepONet10	long	0.0585	0.787	1.27	6.73	1.74	36.3	0.792	10.2	4.9	8.01e-03	Complete
5 ANN10 + T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.910 ms
5 ANN10 + T/P DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.102 ms
5 ANN10 + T/P DeepONet10	long	0.116	0.697	1.86	13	2.64	130	0.954	21.4	11	6.12e-03	Complete
6 T/P/ϕ DeepONet10	base10k	0.896	3.88	2.42	341	53.6	140	0.713	5.19	42.7	8.90e-16	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.016 ms
6 T/P/ϕ DeepONet10	long	0.012	0.283	0.509	1.57	1.55	3.18	0.409	2.64	1.79	4.70e-04	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.807	5.08	5.33	500	48.7	88	0.136	6.07	29.2	8.90e-16	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
7 ANN10 + T/P/ϕ DeepONet10	long	0.112	0.337	1.73	5	2.04	36.9	0.822	11	4.99	5.23e-03	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.